Explain the physics of how we detect neutral hydrogen in space. (2 sentences)

- We detect neutral hydrogen in space by the 21cm radio waves. When an electron and proton in a hydrogen atom flip their spin alignment, the atom transitions to a lower energy state and emits a radio photon with a wavelength of 21cm, which can be detected by radio telescopes

Why do more massive stars live shorter life times? (2 sentences)

- More massive stars have much higher core temperatures and pressures, causing nuclear fusion to occur much faster. As a result, they burn through their hydrogen fuel more quickly, giving them shorter lifetimes.

Explain why the hottest and dimmest stars have no hydrogen lines. (2 sentences)

- In the hottest stars the hydrogen is ionized, so there are no bound electrons to produce hydrogen lines. In cooler stars there is not enough energy to excite electrons to the n=2 level, so the Balmer lines cannot form.

What colour are reflection nebulae and why? (2 sentences)

- Reflection nebulae appear blue in the optical. They look blue because dust scatters shorter wavelengths of light more efficiently than longer wavelengths, reflecting the blue light from nearby stars.

Why are star clusters the ideal place to test our theories of stellar evolution? (2 sentences)

- The stars in star clusters are an ideal place to test our theories of stellar evolution because all the stars in said cluster are born around a very similar time. This allows astronomers to compare how stars of different masses evolve at the same age, making it easier to test stellar evolution.

What is needed to form a molecular cloud and why? (2 sentences)

- Molecular clouds form when gas becomes cold and very dense, allowing hydrogen atoms to combine into molecular hydrogen. Dust is also needed to shield the molecular hydrogen from UV radiation, which keeps the clouds cold and allows molecules to survive.

What colour are HII regions and why? (2 sentences)

- HII regions are pink. This is because ionized hydrogen emits strong H-alpha light when electrons recombine with protons, producing a red emissions line.

What is the main sequence? Explain why 90% of stars are on the main sequence.

- The main sequence is the region of the Hertzsprung–Russell diagram where stars are fusing hydrogen into helium in their cores. About 90% of stars are on the main sequence because hydrogen fusion is the longest and most stable phase of a star's life.

Explain the processes that are responsible for a star expanding into a red giant. (2 or 3 sentences)

- When a star runs out of hydrogen in its core, the core contracts and heats up because fusion stops there. Hydrogen fusion continues in a shell around the inert helium core, producing extra energy that pushes the outer layers outward. This causes the star to expand and cool at the surface, forming a red giant.

Explain the phenomenon of redshift and blueshift. (2 sentences)

- Redshift occurs when a light source is moving away from the observer, causing its spectral lines to shift toward longer wavelengths (the red end of the spectrum). Blueshift occurs when a light source is moving toward the observer, causing its spectral lines to shift toward shorter wavelengths (the blue end of the spectrum).

Midterm 3

What does the colour of a galaxy tell us and why?

- The colour of a galaxy tells us about the ages of its stars and its star formation activity. Blue galaxies have lots of young, hot stars, so they are actively forming new stars.Red galaxies are dominated by older, cooler stars, meaning little recent star formation. This is because hot stars emit more blue light, while cool stars emit more red light, so the overall colour reflects the types of stars in the galaxy.

Describe one prediction of the Big Bang model that has been confirmed by observations.

- One key prediction is that the early Universe was hot and dense and should have left behind a faint glow. This was confirmed by the discovery of the cosmic microwave background, a nearly uniform 2.7 K blackbody radiation observed in all directions.

Describe one reason that was mentioned in class why time machines are unrealistic.

- Time machines are unrealistic because you would have to account for the motion of the Earth, the Sun, and the galaxy through space. If you moved through time without matching that motion, you wouldn't end up on Earth—you'd appear in empty space.

Why do astronomers think that pulsars are neutron stars? (2 sentences)

- Pulsars emit extremely regular pulses with very short periods, which require a very small, dense, rapidly rotating object. This matches the predicted properties of neutron stars, including strong magnetic fields and lighthouse-like beams of radiation.

How do we know that dark matter is necessary for the formation of galaxies?

- Simulations show that with only normal matter, gravity isn't strong enough to form galaxies as early as we observe in the Universe. Dark matter provides extra mass to create gravitational "wells" that let gas collapse and form galaxies. Observations like galaxy rotation curves and structure formation match models only when dark matter is included.

What is the cosmic distance ladder? Name as many "rungs" on the distance ladder as you can, including the rung that calibrates all the others.

- Radar ranging (within the Solar System)
- Stellar parallax (calibrates everything else)
- SpecStars form when gas is cold and dense, allowing gravitational collapse to occur.troscopic parallax / main-sequence fitting
- Cepheid variable stars (period–luminosity relation)
- Type Ia supernovae (standard candles for large distances)
- Redshift (Hubble–Lemaître law) for the largest scales

What was the measurement/discovery that implied that the Sun is not near the centre of the Milky Way?

- The key discovery was mapping the distribution of globular clusters. Harlow Shapley showed that globular clusters are concentrated in a region toward Sagittarius, not around the Sun, which implied the center of the Milky Way is in that direction and that the Sun is off-center.

What is one piece of evidence that spiral galaxies evolve into ellipticals and not the other way around?

- One clear piece of evidence is that we observe galaxy mergers turning spiral galaxies into ellipticals. Colliding spirals lose their disk structure and form a smooth, elliptical system, and we don't see ellipticals turning back into spirals because they lack the gas needed to rebuild disks.

Describe what standard candles are and how they are used in astronomy.

- Standard candles are objects whose true luminosity is known. Astronomers compare how bright they appear to how bright they actually are, and use the inverse-square law to calculate their distance.

Explain the processes that are responsible for a star expanding into a red giant (the first time). (2 or 3 sentences)

- When hydrogen in the core is used up, fusion stops and the core contracts and heats up under gravity. This heats a surrounding shell, where hydrogen shell fusion begins and produces more energy than before. The increased outward pressure causes the outer layers to expand and cool, turning the star into a red giant.

How does the density wave theory explain that dust is seen on the inside of spiral arms while blue stars are seen on the outsides of spiral arms? (2 sentences)

- In density wave theory, gas and dust enter the spiral arm first, where they get compressed and form stars, so dust is seen on the inner edge. The newly formed hot blue stars then move ahead of the arm, so they appear on the outer edge of the spiral arm.

Describe the forces that affect the rate of the expansion and contraction of the Universe.

- The expansion of the Universe is controlled by two main effects. Gravity from matter (including dark matter) pulls things together and acts to slow or reverse the expansion. Dark energy has a repulsive effect that causes space to expand faster, so it drives the current acceleration of the Universe.

Given an example of a universe that is observed to be isotropic and inhomogeneous and a universe that is observed to be anisotropic and homogeneous. (2 sentences)

- An isotropic but inhomogeneous universe would look the same in all directions from a given location, but have clumps of matter (like galaxies and voids) when comparing different locations.
- An anisotropic but homogeneous universe would have matter evenly distributed everywhere, but a preferred direction, so it looks different depending on which way you look.

Why do we think galaxies evolve from spirals to ellipticals?

- We observe galaxy mergers where spiral galaxies collide and lose their disk structure, ending up as smooth, elliptical systems. Ellipticals also have older stars and little gas, suggesting they formed after star formation was used up or disrupted during mergers.

Describe 2 pieces of evidence that black holes exist.

- One piece of evidence is the motions of stars orbiting an invisible, extremely massive object (like at the center of the Milky Way), which implies a black hole. Another is the detection of gravitational waves from black hole mergers, matching predictions from general relativity.

Describe why Cepheid variable stars are a good distance indicator. (2 sentences)

- Cepheid variables have a well-defined period–luminosity relation, so by measuring their pulsation period we can determine their true luminosity. Comparing that to how bright they appear lets us calculate their distance.

Describe the leading theory for the nature of spiral arms in galaxies, and one piece of evidence supporting that theory.

- The leading idea is the density wave theory, where spiral arms are long-lived regions of higher density that rotate through the disk and compress gas, triggering star formation as material passes through. Evidence is that we see dust lanes on the inner edges and young blue stars on the outer edges of spiral arms, exactly as expected from gas entering the wave, forming stars, and then moving ahead.

What are some reasons why astrophysicists think that dark matter is "exotic", ie not like ordinary protons, neutrons and electrons?

- Astrophysicists think dark matter is exotic because it does not emit, absorb, or scatter light, unlike normal matter made of protons and electrons. It also doesn't interact electromagnetically, only through gravity, which is why we detect it indirectly through motions of galaxies and gravitational lensing. Finally, measurements of the early Universe (like the CMB and element abundances) show there isn't enough normal matter to account for the observed mass.

How do we know that dark matter is necessary for the formation of galaxies?

- Observations show that galaxies already existed very early in the Universe, earlier than normal matter alone could collapse to form them. Simulations without dark matter fail to produce galaxies on those timescales, but with dark matter, its gravitational wells let gas collapse quickly and form galaxies.

What is the evidence that pulsars are neutron stars?

- Pulsars produce extremely regular pulses with very short periods, which require a very small, dense, rapidly rotating object. Their properties (size, mass, and strong magnetic fields) match the predictions for neutron stars.